

360° Animation Projects by Students with a Background in Traditional Stop-Motion Techniques

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Figure 1: Examples of 360° stop-motion animation by students

ABSTRACT

Innovations in media technology, exemplified by extended reality, have begun. Hence, the seamless presentation of media, such as animation, film, and games, is progressing at an impressive rate. Integrating these advanced media and experiences into educational environments has excellent potential. Similarly, students who have studied traditional stop-motion animation can discover new forms of expression and enhance their abilities by challenging themselves to incorporate advanced media. This study introduces a unique challenge of stop-motion animation in 360° as a first step toward VR for students who have learned traditional handcrafted object modeling and stop-motion animation. No game engine or VR painting application will be used to ensure that this assignment is free from technical hurdles. Instead, students will complete the project using only an omni-directional camera and standard video editing software.

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1 CONTEXT

Integrating advanced media and experiences into educational environments has excellent potential[Mones 2017]. Educational attempts to connect ceramics and virtual reality (VR) have been practiced and proven effective[Aguilera et al. 2017]. Similarly, students

who have mastered traditional methods of stop-motion expression can expect to practice and develop their skills in advanced media, lead to the revitalization of the entire industry. The project proposed in this paper is part of a first-year master's class in the Department of Animation at Tokyo University of the Arts, which is the only national art university in Japan. Most of the students enrolled in this class have a background in traditional techniques, such as oil painting, handcrafted puppet modeling, and hand-drawn animation. This course is held after the students have completed exercises in hand-drawn animation expression and stop-motion puppet production. The course is positioned to broaden students' perspectives and experiences by introducing them to cutting-edge media. To date, 40% of the students in this course have never viewed virtual reality (VR), and 95% have never been exposed to VR filmmaking.

2 OBJECTIVES

This assignment aims to achieve the following objectives:

- (1) To help novice VR participants understand contemporary VR technology and its application to creative content, with a primary focus on exploring the possibilities of a 360° video expression as an introduction.
- (2) To understand the concept of viewers experiencing 360° of visual freedom, which is not possible for conventional flat images.
- (3) To apply the modeling and stop-motion expressions learned in advance and to develop attractive scenography and video works considering (1) and (2). In this way, students will broaden their perspectives on various forms of expression using cutting-edge media.
- (4) To acquire the skills necessary for filming with an omni-directional camera and editing a 360° video.

3 MATERIALS

The course will consist of three phases: (1) a lecture-style overview of the history of VR and its various applications in movies, games,

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Table 1: Assignment Metadata

Summary	Students will create a 360° animation that combines an immersive visual experience with traditional stop-motion animation.
Learning Outcomes	Through lectures, key concepts of VR and 360° videos are supported, such as video format (Equirectangular), 360° viewing freedom, and viewpoint freedom (3DoF, 6DoF). Through practice, the students will understand the differences from stage production based on a set camera composition in normal stop-motion.
Classification	Stop-motion animation, 360° videos, introduction to VR
Audience	Graduate students at all levels
Dependencies	Basic modeling and stop motion experience, ability to use common video editing software.
Prerequisites	Any form of stop-motion experience, ability to use standard video-editing software.
Strengths	The scope of the assignments goes beyond pure stop-motion animation skill development to help students understand and acquire skills in a variety of video expression contexts that incorporate advanced media.
Weaknesses	Because it is based on live-action filming, it is difficult to immediately develop into 6DoF and interaction, which are inherent features of VR.
Variants	Unlimited production options and modeling techniques.
Assessment	Evaluation will be based on a 30-s 360° animation created by each student and a brief presentation of ideas and concepts.

and other content, (2) a lecture on filming and video editing using a 360° camera, and (3) a week of individual work production. Details of each phase are described as follows:

- (1) First, the introduction will be a lecture-style overview of the history of VR technology and the development of its various representations in movies, games, and other content. Then, several representative works will be presented, and students will be asked to view them wearing a head-mounted display. The works introduced for reference will be classified as three degrees of freedom (3DoF) and six degrees of freedom (6DoF). The students will be asked to experience them and list their differences and characteristics. In addition, the students will be informed that the 360° animations in this course are 3DoF.
- (2) Next, students will learn how omnidirectional cameras work and how to take pictures. Students freely experiment with still photography of sets and other objects created in the stop-motion exercise taken before this course, and they view them on a head-mounted display. Through this, the students will understand the difference in the scope of space recorded using an omnidirectional camera and the concept of composition planned in regular stop motion.

- (3) The assignment is to create and submit a 360° stop-motion animation of approximately 30 s. Based on their experience in (2), students will choose what techniques they would like to develop in their work (e.g., puppet animation and pixilation, including the materials they work with).

The project will emphasize experimentation rather than perfection of the final output and will encourage repeated prototyping and headset viewing to refine the intended visual experience. Although not required, we also encourage the addition of a sound design to the extent possible to further enhance the effect of the production. At the end of the course, students will have to give a presentation on the intent and aims of their work.

4 DISCUSSION

The result of this assignment is a wide variety of 360° animations, focusing on physical and narrative visual experiences and using a variety of approaches (Figure 1). For example, one student created a very elaborate animation in which the occupant's belongings come to life and move in all directions around the room when the occupant is away. Another student drew an animation of a small creature that lives quietly on the ground, unnoticed by humans. For students who are new to VR, this project helps them understand the unique aspects of 360° videos and VR, where the position of the viewpoint has a significant impact on perception. Students who created animations of objects moving over a large area discovered the importance of adjusting the speed of movement for the viewer to easily follow the action. They also understood that precise control over the position and movement of multiple objects and characters on the screen is essential for the viewer to effectively understand the unfolding story. From another aspect, due to time constraints, most students settled on using the actual space to express their ideas.

5 CONCLUSION

In this way, even novice VR students who initially focused on analog methods were able to gain practical experience in 360° production by working on a technically manageable task. This project allowed them to understand the background of a wide range of seamless media expression. In the future, we plan to try to develop this project into a 6DoF production that incorporates a game engine and a production that is possible only with a handmade stage set, which is the best part of stop-motion.

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